MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY : 10548049

(SCALER, RATE)

1. SCOPE

1.1 *This* specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the rate scaler used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548049	Scaler, **Rate**

PACKAGING DATA SHEET

10548049	**Packaging of Rate Scaler**

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548049 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the rate scaler.

3.5 Performance. -For the requirements of 3.5.1 through 3.5.5, logic input and output levels shall be as follows:

Logic O – 0.0 ± 0.5 volt direct current (Vdc) level.
Logic 1 —+3.5 ± 1.0 Vdc level.

3. 5.1 input signals. - Input signals to the rate scaler shall be T1, T2, and T9 as specified in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ± 0.5 volts (V) to +3.5 ± 1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ± 1.0 V to 0.0 *0. 5 V and return to +3.5 V.

3.502 Input Power. - The input power applied between terminals P1-30, P1-31, and P1-32 (+5 V) and terminals P1-24, P1-25, and P1-26 (+5 V return) shall be 5.0 ± 0. 3 Vdc, at 200 milliamperes (mA), regulated; with an alternating current (at) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.3 Re scaled output. - With input logic levels applied sequentially as specified in Table I, the Re scaled output signal at P1-8 shall be as specified in Table I. The out pulse shall have the same characteristics as T1, Figure 1, except the pulse repetition frequency (PRF) shall be as specified in Table L

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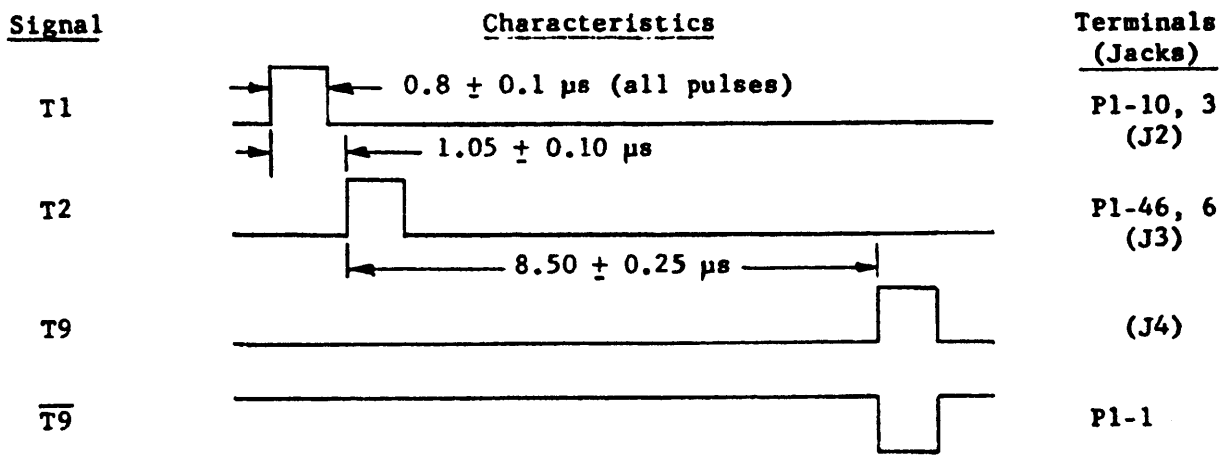


FIGURE 1. INPUT TIMING SIGNALS.

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TABLE I. - Re SCALED OUTPUT SIGNALS

INPUT TERMINALS							
STEP	12	17	38	11	15	13	OUTPUT SIGNAL CHARACTERISTICS
1	1	0	1	0	0	0	Logic O
2	1	0	1	1	0	0	Positive pulse -20 kHz PRF.
3	0	0	1	1	0	0	Logic o
4	1	0	1	1	0	0	Positive pulse -20 kHz PRF.
5	1	0	0	1	0	.0	Logic o
6	1	0	1	0	1	0	Positive pulse -10 kHz PRF.
7	1	0	1	0	0	0	Logic 0
8	1	0	1	0	0	1	Positive pulse -1.25 kHz PRF.

3.5.1 $\overline{R_{up}}$ and $\overline{R_{down}}$ output. - With input logic levels applied sequentially, as specified in Table II, the R up and R down output signals at P1-16 and P1-53 respectively, shall be as specified in Table II. The output pulses ~111'1 have the same characteristics as T 1 as specified in Figure 1 except the polarity shall be as specified in Table II.

TABLE II. - R UP AND R DOWN OUTPUT SIGNALS

INPUT TERMINALS							OUTPUT SIGNAL CHARACTERISTICS	
STEP	12	17	38	11	15	13	PIN-16	PIN-53
1	0	0	0	0	0	0	Logic 1	Logic 0
2	0	1	0	0	0	0	Negative pulse	Positive pulse

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3.5.5 R X G output. - With input logic levels applied sequentially as specified in Table III, the R X G output signal at P1-2 shall be as specified in Table III.

TABLE III. - R X G OUTPUT SIGNALS

	INPUT TERMINALS						
STEP	12	17	38	11	15	13	OUTPUT SIGNAL CHARACTERISTICS
1	0	0	0	1	0	0	T2 as specified in Figure 1.
2	0	0	0	0	0	1	T1 as specified in Figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General, - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4. 5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table IV shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE IV. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ± 0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 200 mA.
2. Pulse generator	Positive pulse 3.5 ± 1.0 V peak-to-peak into 50-ohm load at 20 kHz PRF, 0.8 ± 0.1 us wide. Sync output.
3. Pulse generator	Same as item 2 with external trigger provisions.
4. Pulse generator	Same as item 3.
5. Oscilloscope	Dual channel; rise time of scope plus measuring probe, 10 ns or less; sweep period, 50 ns/cm or greater, input resistance, 8 megohms or greater.
6. Test fixture	As specified in Figure 2. See Table V for test fixture parts list.

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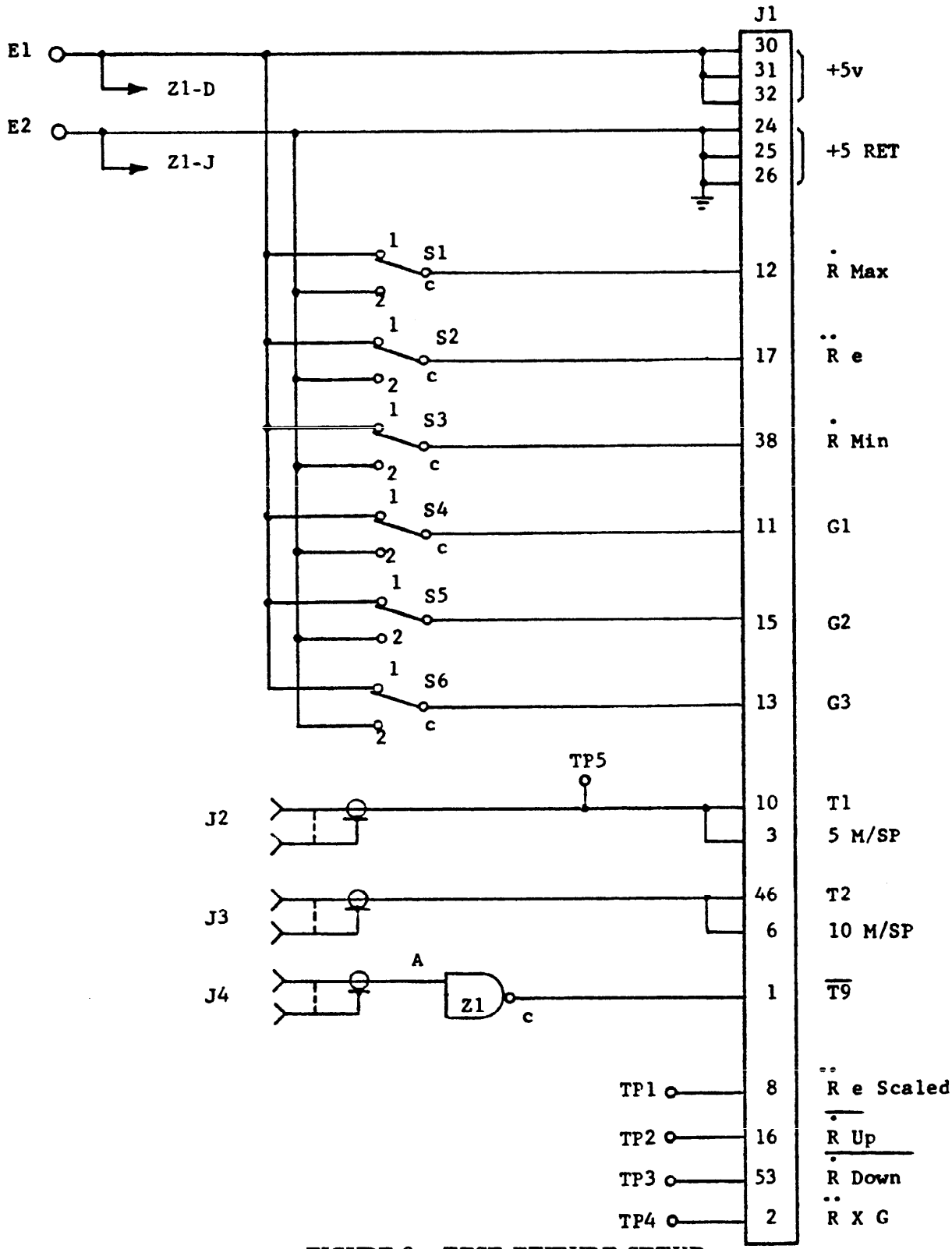


FIGURE 2. TEST FIXTURE SETUP.

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TABLE V. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
El	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co. , Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp. Willow Grove, Pa.
J2 thru J4	Connector, receptacle, UG-25B/U or equivalent.
S1 thru S6	Switch SPDT.
T P1 thru TP5	Tip jack, H.H. Smith 1501 or equivalent.
Z1	Integrated circuit, SG-140-03 Sylvania or equivalent.

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 **Performance.** - The selected rate scaler circuit card assemblies shall be tested **as follows: Set up equipment as shown in Figure 3. Use oscilloscope, item 5, for observation of input and output signals specified in the following paragraphs. When specified, a logic O shall be observed as a 0.0**

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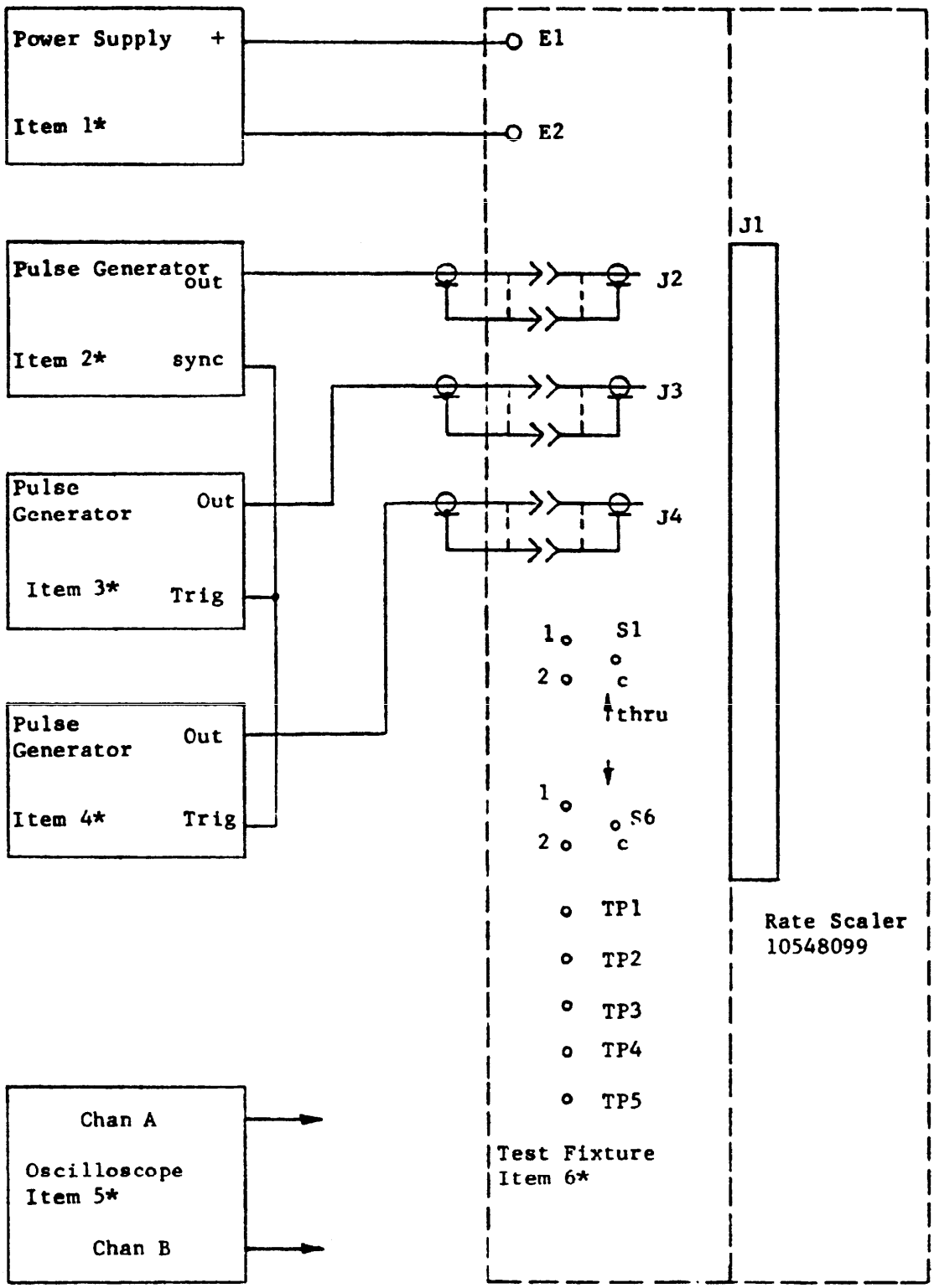


FIGURE 3. RATE SCALER TEST SETUP.

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± 0.5 Vdc level and a logic 1 as a 3.5 ± 1.0 Vdc level. Switches S1 through S6 on test fixture, item 6 are used to supply the logic levels input signals specified in Tables I through III. Switches supply a logic 1 in position 1 and a logic O in position 2.

4.6.3.1 Input signals. - Using pulse generators, items 2, 3 and 4, set up and apply input signals Tl, T2, and T9, as shown in Figure 1 to J2, J3 and J4 on test fixture, item 6.

4.6.3.2 Input power. - Using item 1, apply input power as specified in 3.5.2.

4.6.3.3 Re scaled output. - At TP1 on test fixture, item 6 observe the Re scaled output signal as specified in 3.5.3.

4.6.3.4 R up and R down output. - At TP2 and TP3 on test fixture, item 6, observe that R up and R down output signals respectively are as specified in 3.5.4.

4.6.3.5 X G output. - At RP4 on test fixture, item 6, observe R X G output signal as specified in 3.5.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548049. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. - The rate scaler assembly is used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The rate scaler receives the acceleration or deceleration data (Re signal) and integrates this signal to produce target velocity data (range rate).

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5. 1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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